How to get a new instance of an exported object on each import?

Asked 3 years, 10 months ago Modified 3 years, 10 months ago Viewed 1k times

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I have this file that I'm keeping a INITIAL_VALUE for a form field, that I'm building.

INITIAL_VALUE.js

```
const INITIAL_VALUE = [];

export default INITIAL_VALUE;
```

And the problem is that `INITIAL_VALUE` is an array. A non-primitive, that is handled by reference.

Component1.js

```
import INITIAL_VALUE from "./INITIAL_VALUE";
import React, { useState } from "react";

function Component1(props) {
  const [myState, setMyState] = useState(INITIAL_VALUE);
  const [boolean, setBoolean] = useState(false);

  function handleClick() {
    setMyState(prevState => {
      prevState.push(1);
      return prevState;
    });
    setBoolean(prevState => !prevState);
    props.forceUpdateApp();
  }

  return (
    <React.Fragment>
      <div>This is my state: {JSON.stringify(myState)}</div>
      <button onClick={handleClick}>Modify State Comp1</button>
    </React.Fragment>
  );
}

export default Component1;
```

Component2.js

```
The same as Component1, but it's named Component2 and it has its own file.
```

App.js

```
function App() {
  const [boolean, setBoolean] = useState(false);
```

```
  function forceUpdateApp() {
    setBoolean(prevState => !prevState);
  }

  return (
    <React.Fragment>
      <Component1 forceUpdateApp={forceUpdateApp} />
      <Component1 forceUpdateApp={forceUpdateApp} />
      <Component2 forceUpdateApp={forceUpdateApp} />
    </React.Fragment>
  );
}
```

CodeSandbox

PROBLEM

`Component1.js` and `Component2.js` both import the `INITIAL_VALUE` file. And **I was under the impression that, each one of these imports would get a brand new instance of the** `INITIAL_VALUE` object. But that is not the case as we can see from the GIF below:

enter image description here

QUESTION

Is there a way to keep an array as a initial value living declared and imported from another file and always **get a new reference to it on each import**? Is there another pattern I can use to solve this? Or should I stick with only primitive values and make it `null` instead of `[]` and intialize it in the consumer file?

javascript arrays reactjs object primitive

Share Improve this question Follow edited Aug 1, 2019 at 7:37

asked Aug 1, 2019 at 7:22

cbdeveloper

27.3k • 34 • 149 • 314

1 Answer

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4

> Is there a way to keep an array as a initial value living declared and imported from another file and always get a new reference to it on each import?

No, that's not possible. The top-most level code of a module will run *once*, at most. Here, the top level of `INITIAL_VALUE.js` defines *one* array and exports it, so everything that imports it will have

a reference to that same array.

Easiest tweak would be to export a function which creates the array instead:

```
// makeInitialValue.js
export default () => {
  const INITIAL_VALUE = [];
  // the created array / object can be much more complicated, if you wish
  return INITIAL_VALUE;
};
```

and then

```
import makeInitialValue from "./makeInitialValue";
function Component1(props) {
  const INITIAL_VALUE = makeInitialValue();
  const [myState, setMyState] = useState(INITIAL_VALUE);
```

In the simplified case that you just need an empty array, it would be easier just to define it when you pass it to `useState`.

All that said, it would be much better to fix your code so that it does not mutate the existing state. Change

```
setMyState(prevState => {
  prevState.push(1);
  return prevState;
});
```

to

```
setMyState(prevState => {
  return [...prevState, 1];
});
```

That way, even if all components and component instances start out with the same array, it won't cause problems, because the array will never be mutated.

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edited Aug 1, 2019 at 7:34

answered Aug 1, 2019 at 7:26

CertainPerformance

353k • 51 • 301 • 313

Thanks for your answer. I just tried that, but what happens is that `Component2` gets a new instance. But both `Component1` end up sharing the same instance. Is there a solution to this? – cbdeveloper Aug 1, 2019 at 7:28

There is no need to invoke `INITIAL_VALUE` function, it can be passed as-is to `useState` and it will be lazily initialized: reactjs.org/docs/hooks-reference.html#lazy-initial-state – haim770 Aug 1, 2019 at 7:29

My real case I have like 5 possible initial values that are based on some form field `type`. Some of the types should have empty `array`, other will have empty `string`, others are `false`, etc. – cbdeveloper Aug 1, 2019 at 7:30

My bad, I was invoking the function at top-level. If I invoke it inside the component is all good and every one gets its own instance! Thanks!! – cbdeveloper Aug 1, 2019 at 7:31

@cbdev420 Keep in mind that this is a *bit* of an X/Y problem, for this particular situation with React - it would be better never to mutate the existing state at all. Better to create a new array every time `setMyState` is called – CertainPerformance Aug 1, 2019 at 7:35